

ENERSCRIPT | GUIDE: RAJA KUMAR

MudraTrack · GST Invoicing & Job-Card Management ERP

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Project Overview



MudraTrack

ERP and billing system for print shops, digital packaging studios, and banner manufacturers. Integrates job-card workflows with GST-compliant invoicing and role-based access control.



Key Features

Prepaid wallet (Razorpay), automated WhatsApp & email notifications, and an AI assistant (Google Gemini) for business reporting.

Technologies — Core Stack



Frontend

React.js (v18), Redux Toolkit, Apollo Client



UI & Motion

Material UI, Framer Motion



Backend

NestJS, TypeScript, GraphQL, Prisma ORM



Services

Node-cron, MongoDB, Razorpay integrations,
Google Gemini AI

Frontend Work Completed

Multi-Company Switching

Automatic Apollo cache sync across tenant contexts for seamless switching.

Mobile Responsiveness

Dashboard, navigation, and forms optimized for all screen sizes.

Localization

Marathi support added; Hindi locale corrected. Full i18n across UI strings.

Wallet UI

Razorpay Checkout SDK integrated for prepaid wallet recharges.

Backend Work Completed

Developed a dynamic pricing engine and volume-based slab discount calculator to produce deterministic invoice totals. Implemented job-card workflows, GST-compliant invoice generation, and automated notifications using Node-cron.

Stack: NestJS + GraphQL + Prisma ORM + MongoDB. Emphasis on correctness for financial calculations and robust business rules enforcement.

Key Learnings

Monorepo & CI/CD

Full-stack ERP development with monorepo patterns and automated pipelines.

Payment Security

Secure payment gateway flows with webhook signature validation and constant-time comparisons.

Localization

Complete Marathi & Hindi support, including locale testing and fallback strategies.

AI Integration

Google Gemini for natural-language business intelligence and reporting.

Scientific Computing & Mathematical Modeling

1

Piecewise Slab Pricing Curve

Model: $P(x) = \sum_{i=1}^n p_i \cdot I_{[s_{i-1}, s_i)}(x)$. Volume discounts modeled as a piecewise step function where s_i are quantity boundaries and p_i are rates. Guarantees deterministic totals and avoids floating-point drift by using integer units and exact arithmetic.

2

Cryptographic Webhook Validation

Model: $\text{HMAC}(K, m) = H((K' \oplus \text{opad}) \parallel H((K' \oplus \text{ipad}) \parallel m))$. Razorpay webhook signatures validated with HMAC-SHA256. Implement constant-time comparison to mitigate timing side-channels; validation routines run in $O(1)$ time relative to input length checks.

3

Statistical NLP Vector Mapping

Google Gemini maps business text to embeddings for reports and anomaly detection. Use vector similarity for intent classification and structured extraction to augment BI queries.

Impact & Summary

Impact — Streamlined print & packaging workflows with integrated GST-compliant invoicing. Improved UX via responsive design and Marathi/Hindi localization.

Summary — Secure automated financial transactions (Razorpay) with cryptographic webhook validation. AI-driven insights (Google Gemini) enable data-backed business decisions and reporting automation.

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Organization

Enerscript — Platform, resources, and real-world environment for deployment.

Peers & Mentors

Thanks to peers, mentors, and the open-source community for contributions and support.

MUDRATRACK · ERP PRESENTATION

Thank You

Questions? Let's Discuss!